

Late-layer readouts break

after duplex fine-tuning



spoken query



last-layer probe

held-out math $.93 \rightarrow .37$



verbalized $p(\text{True})$

collapses on audio input

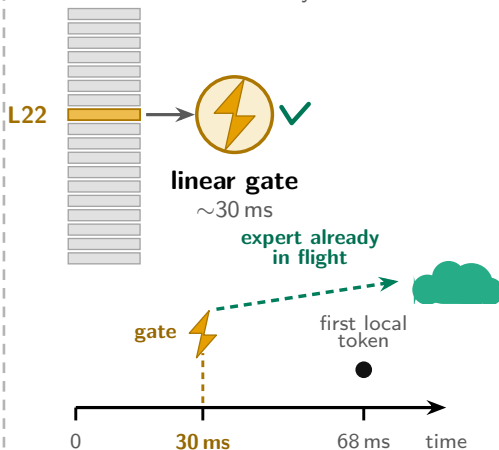


decode-time entropy

arrives after the first token

Mid-network, and early

one linear read at layer 22 of 36

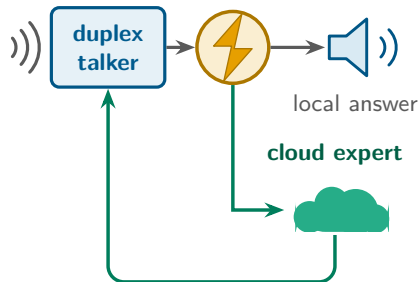


decided before the first token

LOPO $.93$ · text $\rightarrow$ speech $.86$
streaming end-of-turn AUC $.887$

Same talker speaks it

session and voice stay local



the local voice speaks it

live spoken session

heard acc. $.422 \rightarrow .633$
at 57% escalation

frozen, label-free transfer

Speech TriviaQA $.664 \rightarrow .860$
vs. $.755$ official offline